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Proof-of-concept of host attribution of antimicrobial resistance genes using wastewater Hi-C metagenome sequencing

Kongyang Zhu[a,b], Ayaaz Amirali[a], Benjamin Auch[c], Kristina M. Babler[a,d], Pratim Biswas[a], Kate Bowie[e], Shruti Choudhary[a], Benjamin B. Currall[f], George S. Grills[f], Hannah G. Healy[g], Ivan Liachko[c], Alexander G. Lucaci[h], Christopher E. Mason[h,i,j], Mark Sharkey[k], Oona Shigeno Risse-Adams[h], Bhavarth S. Shukla[k], Zach Sisson[c], Mario Stevenson[k], Sion L. Williams[f], Alessandro Zulli[e,l], Jordan Peccia[e] and Helena M. Solo-Gabriele[a,*]

[a] Department of Chemical, Environmental, and Materials Engineering, University of Miami, Coral Gables, FL 33146, USA
[b] Department of Industrial and Systems Engineering, University of Miami, Coral Gables, FL 33146, USA
[c] Phase Genomics, Seattle, WA 98109, USA
[d] Current address: Utah Public Health Laboratory, Taylorsville, UT 84129, USA
[e] Department of Chemical and Environmental Engineering, Yale University, New Haven, CT 06511, USA
[f] Sylvester Comprehensive Cancer Center, University of Miami Miller School of Medicine, Miami, FL 33136, USA
[g] Department of Environmental Health, T.H. Chan School of Public Health, Harvard University, Boston, MA 02115, USA
[h] Department of Systems and Computational Biomedicine, Weill Cornell Medicine, New York, NY 10065, USA
[i] The HRH Prince Alwaleed Bin Talal Bin Abdulaziz Alsaud Institute for Computational Biomedicine, Weill Cornell Medicine, New York, NY 10021, USA
[j] The WorldQuant Initiative for Quantitative Prediction, Weill Cornell Medicine, New York, NY 10021, USA
[k] Department of Medicine, University of Miami Miller School of Medicine, Miami, FL 33136, USA
[l] Department of Civil and Environmental Engineering, Stanford University, Stanford, CA 94301, USA
*Corresponding author. E-mail: hmsolo@miami.edu

HS-G, 0000-0003-3390-3823

ABSTRACT

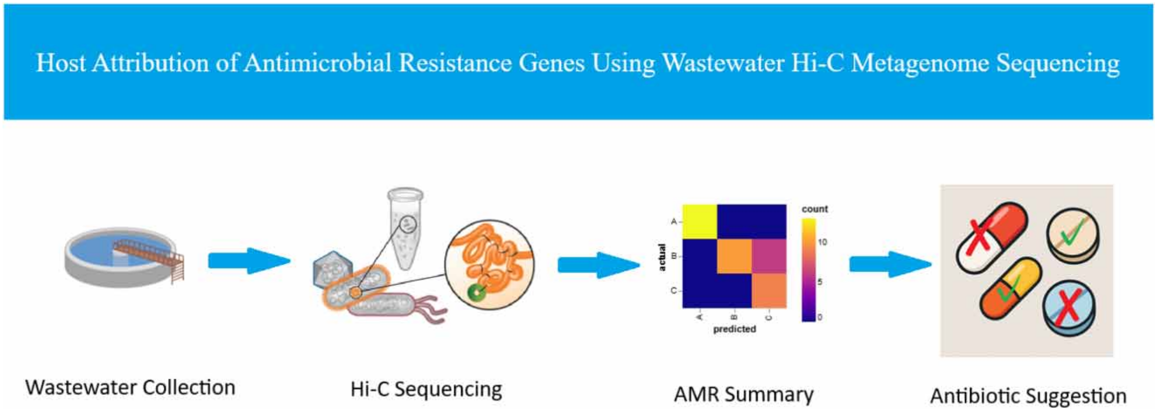
The proliferation of antimicrobial resistance genes (ARGs) poses public health risks globally, with wastewater treatment plants (WWTPs) serving as dissemination hubs for horizontal gene transfer. In this study, we evaluated the potential of applying Hi-C sequencing coupled with metagenomic bioinformatics for surveillance of ARGs and other microbial fitness traits using samples from WWTPs. Hi-C sequencing has the advantage over other molecular approaches by directly associating genes conveying fitness to their host microbe, plus to their element type (in plasmids, phages, or within the core genome of its host microbe). Results from Hi-C analyses confirm results from more laborious approaches by showing that aminoglycoside resistance is disseminated by plasmids. Mercury resistance was found in *Zoogloea* bacteria. Resistance genes to quaternary ammonium compounds were found within bacteriophages. Results from this study provide proof-of-concept for the potential value of Hi-C metagenome sequencing in wastewater attribution studies by illustrating the breadth of information that can be obtained about the microbial community, the exchange of genes, and their interconnections. We believe that with further development, Hi-C sequencing can be integrated into routine monitoring of wastewater for the purpose of providing near-real-time information about the dissemination of fitness traits, including ARGs.

Key words: aminoglycoside, antimicrobial resistance genes (ARG), high-throughput chromosome conformation capture (Hi-C), quaternary ammonium compounds (QACs), stress genes, virulence genes

HIGHLIGHTS

- Monthly wastewater samples analyzed via Hi-C sequencing for ARG attribution.
- Results indicate the origin of ARG as within the genome, plasmid, or phage-associated.
- Aminoglycoside resistance is found primarily in plasmids.
- Mercury resistance is found in plasmids of *Zoogloea* bacteria.
- Future applications may include identification of antibiotics to treat circulating infections.

GRAPHICAL ABSTRACT



INTRODUCTION

Microbes carry fitness traits that enable their survival. Among the fitness traits, antimicrobial resistance genes (ARGs) pose one of the most pressing threats to global public health (Laxminarayan *et al.* 2013), driven by the widespread use and misuse of antibiotics, resulting in the spread of bacterial infections resistant to antibiotics (Aminov 2011). Additional fitness traits include stress genes which impart resistance to environmental stressors such as elevated metals, disinfectants, heat or acidic conditions, and virulence genes which increase the potency of microorganisms to cause disease through enhanced mechanisms to infect the host or to evade the host immune system.

In this article, the biological location of fitness genes were separated into three different gene elements types, one of which is referred to as 'genomic' and considered to be low mobility because it is found within the core genome and not located in an integrated plasmid, versus two additional mobile gene elements (MGEs) which are considered to have high mobility, 'plasmids' (either integrated or extrachromosomal), and gene elements transmitted by bacteriophages called 'phage-associated'. Generally, the greatest concern for environmental transmission of ARGs tends to be among the MGEs, given the ease with which genetic information is transferred between microbes. According to the National Center for Biotechnology Information (NCBI) Pathogen Detection Microbial Browser for Identification of Genetic and Genomic Elements (MicroBIGG-E) as of 24 October 2025, the top MGEs in the United States corresponded to three ARG families: beta-lactam with 4,312,334 reports, aminoglycoside with 2,944,129 reports, and tetracycline with 1,588,440 reports. The MGE mercury stress gene, a fitness trait that provides resistance to elevated mercury levels, was in fourth place with 1,453,227 reports (NCBI 2025b).

Microbes, in particular bacteria, can acquire fitness traits, and more specifically antimicrobial resistance, through the accumulation of *de novo* mutations (vertical gene transfer), or uptake and incorporation of chromosomal DNA or extrachromosomal plasmids via horizontal gene transfer (HGT), of which the latter is considered to be the most important factor in the current pandemic of antimicrobial resistance (Von Wintersdorff *et al.* 2016). Wastewater treatment plants (WWTPs) are considered as potential hotspots for ARGs and other fitness trait dissemination via HGT (Rizzo *et al.* 2013). WWTPs aggregate inputs from domestic, industrial, and hospital wastewater, assimilating diverse microbial communities, creating hotspots for the persistence and potential HGT activities of MGEs (Von Wintersdorff *et al.* 2016; Jian *et al.* 2021). Previous studies have reviewed the mechanisms of HGT of ARGs, which can be observed in WWTPs. Among these mechanisms, plasmid molecules can be transferred via conjugation, a process that requires cell-to-cell contact, transformation, a process where the plasmids released from dead cells are taken up intact by some bacteria, or transduction, where the plasmids are transmitted by accidentally being packaged into bacteriophage head coats (Davison 1999; Slonczewski & Foster 2014; Von Wintersdorff *et al.* 2016). Notably, both intraspecific (gene transfer between same species) and interspecific (gene transfer between two different species) HGT activities via plasmids have been observed in laboratory sewage experiments (Mancini *et al.* 1987) and within mesocosm studies at full-scale WWTPs (Marcinek *et al.* 1998). Studies have also shown the HGT activity of bacteriophages facilitating the dissemination of antimicrobial resistance through transduction with or without the help of plasmids (Calero-Cáceres *et al.* 2014; Marti *et al.* 2014; Slonczewski & Foster 2014; Risely *et al.* 2024).

The observation of antimicrobial resistance genes and other fitness traits has been facilitated through advances in high-throughput sequencing (HTS), especially High-throughput Chromosome Conformation Capture (Hi-C) sequencing. Unlike traditional culture-dependent methods, metagenomics study captures the entirety of the microbial ecosystem, allowing for the detection of both culturable and unculturable organisms and ARGs plus genes carrying other fitness traits (Slizovskiy *et al.* 2020; Yang *et al.* 2021; Royer *et al.* 2024). A conventional HTS method is limited in that it does not provide information about the relationships between extrachromosomal plasmids or phage-associated factors and their bacteria host, especially in the environmental samples due to its microbial diversity. Thus, a wide myriad of ARGs, vectors, and their corresponding host combinations are difficult to determine in their entirety using conventional HTS approaches. Hi-C sequencing provides the opportunity to study the entire population and interconnections of ARGs with their hosts bacteria through proximity ligation of MGEs vectors to the chromosomal DNA of host microorganisms prior to extraction, allowing for the metagenomic sequencing of the ligated DNA pool (Marbouty *et al.* 2017, 2021; Stalder *et al.* 2019). The sequencing approach is based on chromosome conformation capture, which chemically cross-links spatially adjacent DNA fragments within intact cells through the action of formaldehyde (Calderón-Franco *et al.* 2023; Wu *et al.* 2023). Thus, by preserving these spatial associations, Hi-C sequencing results provide a powerful means to reconstruct the MGEs–vector–host microbe interactions for the entire sample analyzed, and enables the study of connections between ARGs and their microbial hosts in complex environments like wastewater samples.

The objective of this study was to illustrate the utility of Hi-C sequencing, specifically providing pilot data that illustrate the potential of the technology for analyzing wastewater. For this study, wastewater samples were collected monthly over a year-long period. Using Hi-C sequencing and metagenomic analysis, we identified MGEs (carrying fitness traits of ARGs, stress genes, and virulence genes) in wastewater samples by gene element (genomic, plasmid, or phage-associated) and linked them to their microbial hosts at the genus and species levels; we evaluated the prevalence and diversity of ARGs targeting specific classes of antibiotics or stress types; we identified possible ARGs dissemination through HGT; and we provide suggestions towards the application of the results for local healthcare systems and describe future research based on the information collected. By using cutting-edge sequencing technologies to analyze wastewater, this research provides a proof-of-concept of how wastewater can be used to illuminate the mechanisms underlying ARG persistence and dissemination in urban WWTPs using Hi-C sequencing methods. Insights gained from wastewater can be potentially used towards public health strategies aimed at mitigating the spread of ARGs and other MGEs in both environmental and clinical settings.

METHODS

Wastewater sample collection

All wastewater was collected from a regional WWTP in Miami-Dade County (Central District WWTP, 830,000 service population) (Tierney *et al.* 2024). Wastewater samples collected at the Central District WWTP corresponded to 24-h (midnight to midnight) flow-weighted composite samples using a refrigerated autosampler (HACH AS950 fitted with an IO9000 for flow proportional sampling). Samples were collected monthly in 0.5 L sterile bottles from January 2023 through January 2024 (13 samples collected in total). Upon collection, samples were spiked with a process control (OC43 heat inactivated to a concentration of 10^6/L) and then split to generate three filters by vacuum filtration (40 mL filtration volume each) using Pall GN-6 filters (47 mm diameter, 0.45 μm pore size) per protocol developed by Babler *et al.* (2023, 2025).

The first filter was processed by the same team in a SARS-CoV-2 Wastewater-Based Surveillance project described by Amirali *et al.* (2025). The second and third filters were used for host attribution analysis via Hi-C sequencing. These filters were processed by first scraping the sediment collected from the top of both filters and placing this scraping into 1,000 μL of PGShield (Phase Genomics 2023). This preservative differs from other nucleic acid preservatives such as Zymo DNA/RNA Shield (used for first filter) as it keeps the bacterial cells intact which is necessary for host attribution. These samples were shipped to Phase Genomics Inc. (Seattle, WA, USA) for further processing for Hi-C sequencing and metagenomic analysis.

Hi-C metagenomics process

The ProxiMeta™ Hi-C Kit (Protocol v4.0, Lieberman-Aiden *et al.* 2009) was used to process the sample concentrates. The Hi-C process begins with resuspension in water and then centrifugation at 1,000*g* for 10 min (Wu *et al.* 2023). The sample was then split with one split subjected to shotgun sequencing (not cross-linked) and the other subjected to Hi-C sequencing (cross-linked). For the split subjected to Hi-C sequencing, formaldehyde was applied to the supernatant of the centrifuged sample to a final concentration of 1% (v/v) to generate covalent links between spatially adjacent genetic segments (Calderón-Franco *et al.* 2023). Processed samples were incubated at room temperature for 20 min and then incubated in glycine (125 mM, 1%w/v) for another 20 min at room temperature to halt the crosslinking reaction. The cross-linked DNA was then extracted by bead beating and a detergent-based lysis reagent that is part of the ProxiMeta™ Hi-C Kit. Upon extraction, the sample was fragmented with restriction enzymes, and subjected to proximity ligation, where spatially linked DNA fragments were joined together (Wu *et al.* 2023). After reverse crosslinking, the DNA was purified and Hi-C junctions were bound to streptavidin beads. The streptavidin-bound Hi-C DNA was then washed to remove unbound DNA and prepared for paired-end deep sequencing using the ProxiMeta Library Preparation reagents.

The split that was not cross-linked and subjected to shotgun sequencing was extracted using the ZymoBiomics DNA Miniprep Kit. Shotgun libraries were prepared using the Watchmaker DNA Library Prep Kit with Fragmentation. Shotgun and Hi-C libraries were sequenced on an Illumina NovaSeq X using 2×150 sequencing, targeting 200M read-pairs for the shotgun library and 100M read-pairs for the Hi-C library for each sample. The sequencing data can be accessed from the SRA database under project PRJNA1272037. Shotgun reads were processed by fastp (version 0.20.1) using default parameters for adapter trimming and read quality filtering, followed by assembly with MEGAHIT (version 1.2.9) using default parameters (Li *et al.* 2015, 2016; Chen *et al.* 2018). Hi-C metagenomic data were aligned and assembled into metagenome-assembled genomes (MAGs) through the ProxiMeta™ Metagenome Deconvolution Platform (Phase Genomics, Seattle, WA). Additional details of Hi-C libraries preparation, sequencing, and data processing are explained in Burton *et al.* (2014), Stalder *et al.* (2019), and Press *et al.* (2017).

Statistical analysis

The original shotgun reads, Hi-C reads, contig FASTA files, alignment files, assembled MAGs library, mobile elements (plasmids, phages, and integrative conjugative elements (ICEs)) data and fitness trait genes data were obtained from the ProxiMeta™ platform (Phase Genomics 2025b) and cross-checked against the Genome Taxonomy Database (GTDB 2025) and the NCBI databases (NCBI 2025a) with SnapGene (version 7.1.1) (SnapGene 2025). MGE annotation from the metagenomic assembly was performed using AMRFinderPlus (version 3.10.5) with the –plus option enabled, and default parameters in ProxiMeta Explorer. Fitness trait gene annotations were cross-referenced with MAG taxonomic assignments and the host-phage and host-plasmid association matrices generated by ProxiMeta to determine the element type (i.e., genomic, plasmid, or phage-associated) of the fitness trait. A binary matrix was created to track the presence of fitness traits in each MAG and classify their element type accordingly. Fitness traits found in wastewater samples were classified into three major gene types: (1) antibiotic resistance genes (ARG shown as 'AMR' type in MicroBIGG-E (NCBI 2025b) and ProxiMeta™ Explorer Platform (Phase Genomics 2025a)); the ARGs included resistance to antibacterials used to treat bacterial infections like beta-lactams, macrolides, quinolones, tetracyclines, or aminoglycosides, (2) stress genes ('stress' for short in the following, shown as 'STRESS' type in MicroBIGG-E (NCBI 2025b) and ProxiMeta™ Explorer Platform (Phase Genomics 2025a)); the stress genes refer to resistance to non-pharmaceutical heavy metal elements or compounds, biocides, heat or acid; and (3) virulence factor encoding genes ('virulence' for short in the following, and listed in MicroBIGG-E (NCBI 2025b) and ProxiMeta™ Explorer Platform (Phase Genomics 2025a)); the virulence genes are also known as pathogenicity factors or effectors which cause damage to the host.

The tabular data of 11 successful lab runs from 13 processed samples in different months were aggregated and arranged with Microsoft Excel, and the statistical analysis was processed with SAS (version 9.4_M7). Two of the 13 samples failed to meet the minimum requirements for sequencing quality. Fitness trait reads for host attribution study were counted only for metagenome-assembled genomes (MAGs) corresponding to 70% completeness (estimate of the percentage of the genome that is represented in the assembly) for a medium quality (Nayfach *et al.* 2021) and contamination rate (estimate of the percentage of the genome that is derived from contaminating sequences) of less than 10%. Completeness and contamination rates were computed through CheckM, CheckV, and the NCBI Plasmid Database (NCBI 2025c). Fitness trait reads (two of the 13) connected to MAGs quality failure criteria from either of the computational methods were omitted from the host attribution analysis.

High-frequency fitness traits were defined as those with a read count higher than the median values among all fitness trait read frequencies at a 99% confidence level. Results from the Shapiro–Wilk test indicated that the frequency data were not normally or lognormally; thus, a Wilcoxon Signed-Rank test was performed to determine the 99% confidence limits for the median point.

The fitness trait frequency distribution and gene element type summary were visualized for further analysis by SAS (version 9.4_M7). Aggregated heatmaps for data visualization of fitness trait–host connections were generated using SAS and the

ProxiMeta™ Explorer Platform (Phase Genomics 2025a) provided by Phase Genomics Inc. Visualizations were prepared from the phylum to species and lower taxonomic levels within ProxiMeta™ Explorer, based on Kanaries PyGWalker (https://github.com/Kanaries/pygwalker).

RESULTS AND DISCUSSION

After screening the 13 samples collected, 11 samples were successfully aligned and assembled into a MAG library. Once assembled, 2,816 copies of fitness traits were successfully identified, of which 554 were related to a qualified possible bacterial host. Evaluating results in time series showed that the sum of fitness trait reads is highest in early summer. The values ranged from the highest 113 total reads, or 0.070 reads per million base pairs (Mbp) in June 2023, to the lowest value of five total reads or 0.0029 reads per Mbp in September 2023 (Supplementary Figure S1(a) and Table S2). No clear monthly trend could be identified through the year; this agrees with the finding in a previous study of the same WWTPs during 2021 and 2022 (Tierney *et al.* 2024).

The 554 fitness traits associated with a possible bacterial host were separated into the three major types and 23 gene classes (Figure 1), of which 52.9% ($n = 293$) of these genes were ARGs (shown by red bars), 46.2% ($n = 256$) were stress genes (shown by green bars) and 0.90% ($n = 5$) were virulence factor encoding genes (shown by the blue bar). The result of the Wilcoxon

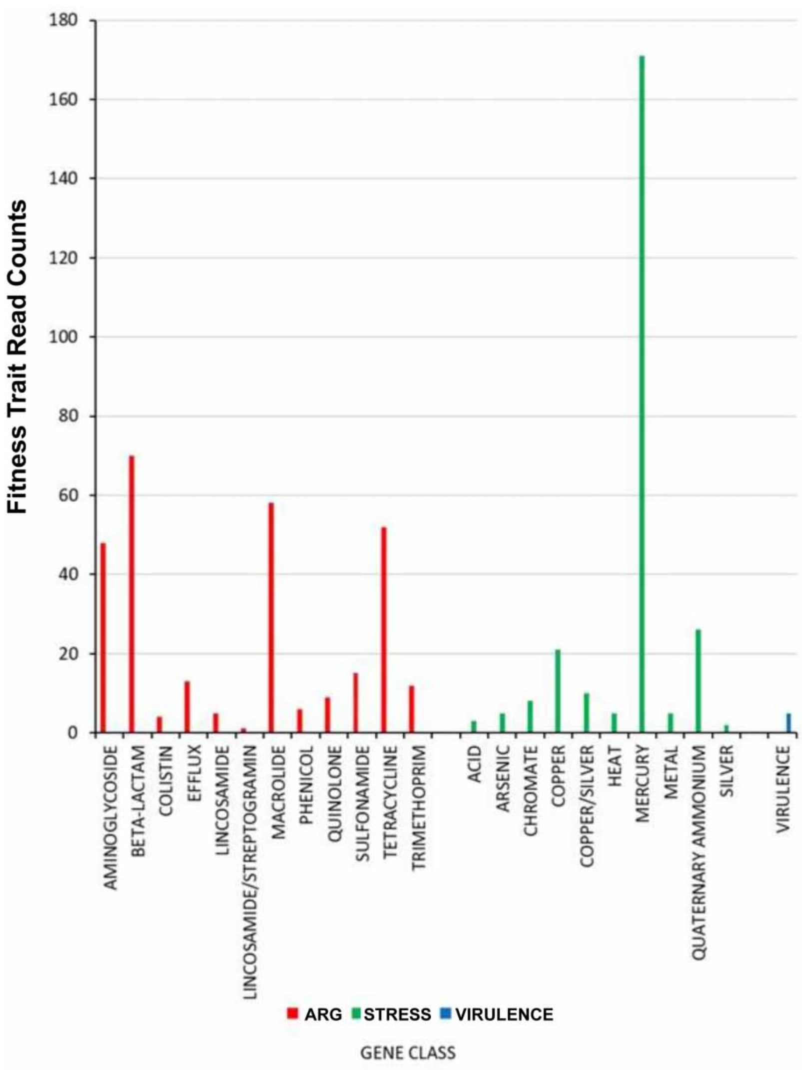


Figure 1 | Fitness trait read counts distribution by element type (ARG, stress, virulence) and gene class.

Signed-Rank test suggested that fitness trait reads with a frequency higher than 48 can be considered as high-frequency fitness traits in this study, which includes aminoglycoside (frequency, $f = 48$), beta-lactam ($f = 70$), macrolides ($f = 58$), tetracycline ($f = 52$), and mercury ($f = 171$) (Supplementary Table S2). This result is consistent with the five most frequently detected fitness traits according to MicroBIGG-E (Supplementary Table S3, NCBI 2025b). The monthly read counts of these five fitness traits show similar trends to monthly total fitness trait read counts, except for aminoglycoside (Supplementary Figure S1(b)).

Fitness trait distribution by gene element type

The 554 fitness trait genes were found in either a host genome ($n = 289$) or a gene vector ($n = 265$), with the fitness trait connected to a vector either showing up in a plasmid vector ($n = 263$) or bacteriophage ($n = 2$) (Figure 2, Supplementary Table S2). ARG genes (beta-lactam, macrolides, tetracycline, etc.) tend to be located inside the host genome, while stress genes (mercury, copper, quaternary ammonium, etc.) tend to be located inside a vector. Five ARG genes (trimethoprim, sulfonamide, quinolone, phenicol, and aminoglycoside) showed up in vectors more so than in host core genomes. Aminoglycoside was the only high-frequency fitness trait among these five. Quaternary ammonium was the only fitness trait observed to have been carried by a bacteriophage in this study. Virulence genes were detected in host genomes only.

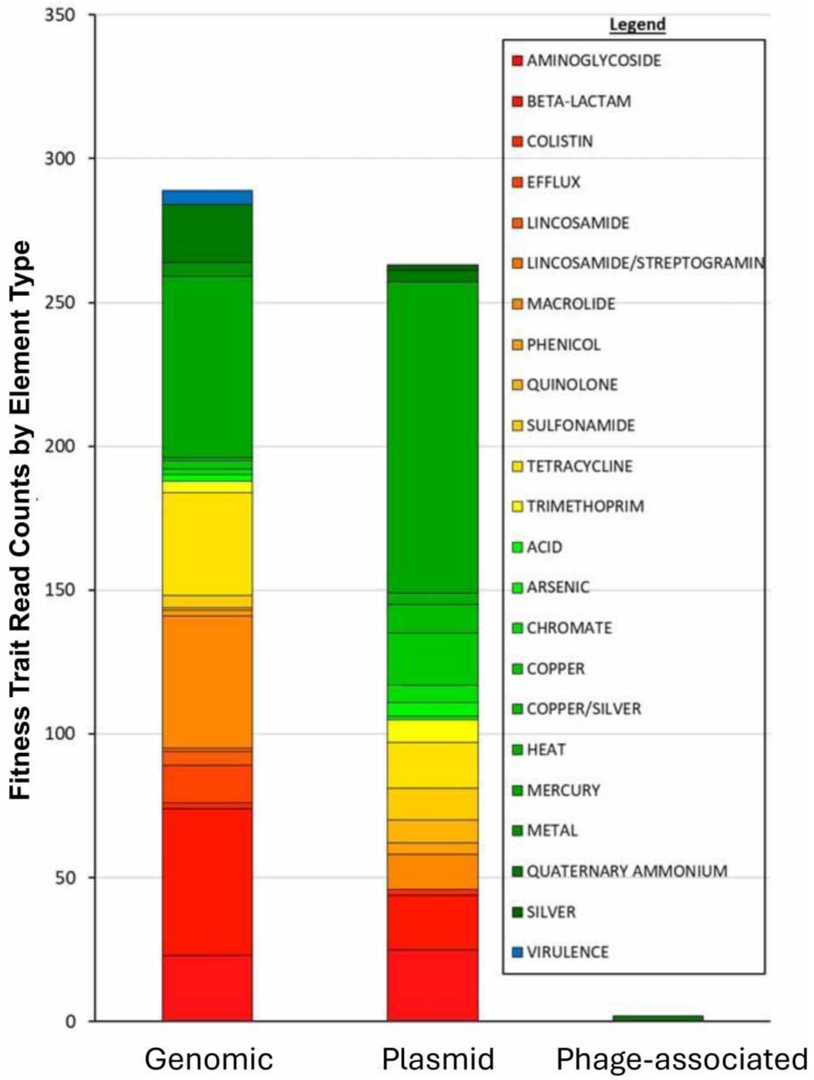


Figure 2 | The fitness trait class distribution by its element type. ARGs are shown in red, orange, or yellow colors, stress genes are shown in green colors, and virulence genes are shown in blue.

Fitness trait host attribution

Among all 554 fitness traits associated with a possible bacterial host, 238 of the host genomes had a bin of completeness of 70% or higher calculated through CheckM (Ecogenomics 2025b). Seventeen of the host bins showed a 10% or higher marker gene overrepresentation, which is an estimate of the percentage of the genome that is derived from contaminating sequences (Phase Genomics 2025a). After removal of the contaminating genes, a total of 221 fitness trait–host interactions were counted in the current study. The subsequent analysis of fitness trait–host interactions were performed at different taxonomic levels, from phylum to species, by the fitness trait element type. As taxonomic levels go lower, the total number of associations decrease due to the quality of alignment.

Starting with the phylum level (Supplementary Figure S2(a)), the mercury resistance gene family in *Pseudomonadota* showed the highest frequency either in host genomes ($n = 26$) or carried by a plasmid ($n = 42$) as well as the tetracycline-resistance gene family ($n = 25$), mostly in host genomes ($n = 19$). Most of these *Pseudomonadota* are composed of *Burkholderiales*, *Methylococcales*, *Pseudomonadales*, and *Enterobacterales* at the order level (Supplementary Figure S2(c)). The most significant portion of mercury resistance gene host was identified as *Zoogloea* sp. ($n = 29$, Supplementary Figure S2(e)). More noticeably, plasmids containing aminoglycoside resistance genes were widely discovered at the order and lower taxonomic levels (Supplementary Figure S2(c)–S2(f)); this might explain why aminoglycoside resistance genes were more consistent than other ARGs throughout the year in WWTP samples.

Only a small fraction of MAGs could have their taxonomy resolved to the species level (Figure 3). The most noticeable interaction was the tetracycline-resistance gene family located in the genome of JACKOQ01 sp024234215, according to GTDB (GTDB 2025) from samples of January, February, March, April, and December 2023 and January 2024, which includes *tmexC1* (30%), *tmexD1* (35%), and *toprJ1* (35%) ARGs (Figure S3). JACKOQ01 sp024234215 species is a *Pseudomonadales* bacterium not recorded in NCBI. Human gut microbes like *Bacteroides graminisolvens*, *Macellibacteroides fermentans*, *Phocaeicola vulgatus*, and *Shewanella baltica* were shown to carry multiple classes of ARG genes within their genome. For example, *B. graminisolvens* was shown to carry lincosamide-, macrolide-, and tetracycline-resistance genes. Notably, here we spotted multiple opportunistic human pathogenic species linked to ARGs. Some carried the ARGs in the host genome such as *Mycobacterium mageritense* carrying macrolide-resistance and *Laribacter hongkongensis* carrying beta-lactam resistance (which is consistent with the findings of Teng *et al.* (2021)). Other opportunistic pathogens were associated with ARGs through plasmids such as in *Shewanella xiamenensis* carrying beta-lactam resistance and *Streptococcus suis* carrying macrolide resistance (which is consistent with the findings of Uruén *et al.* (2022)).

Some interactions were not included in the heatmaps due to typesetting limitations, but were still notable: all virulence genes were found in the host genome of *Tolumonas* from one single sample. These include the aerobactin synthase gene family (*iucA*, *iucB*, *iucC*, *iucD*), which is a bacterial iron chelating agent (siderophore), and ferric aerobactin receptor *iutA*, which was found to have a carbohydrate-binding property and to be important for colonization in *E. coli* (Landgraf *et al.* 2012). These relationships were not recorded on the NCBI database (NCBI 2025b). The only fitness trait class carried by bacteriophage was quaternary ammonium resistance genes, found in *Aureliella*.

Comparison of results to reported MGEs

The Hi-C analysis based on local wastewater agrees with the nationwide MGE reports in overall frequency. Three of the top ARGs discovered in this study (beta-lactam, tetracycline, and aminoglycoside) were also the top three MGEs recorded in the NCBI database (beta-lactam, aminoglycoside, and tetracycline) based on whole genome sequencing (Figure 1, Supplementary Table S2). However, some classes of ARG were found in plasmid vectors more than in host genomes (trimethoprim, sulfonamide, quinolone, phenicol, and aminoglycoside), of which aminoglycoside was the only high-frequency fitness trait both in our study and in the NCBI database. In this study, aminoglycoside resistance could be found in plasmids related inside *Acinetobacter*, *Novosphingobium*, *Rivihabitans*, and *Tolumonas*, where *Acinetobacter* is an opportunistic pathogen associated with serious hospital acquired infections and its resistance to aminoglycoside has been recorded (Rizk & Abou El-Khier 2019); the only species confirmed with plasmids containing aminoglycoside resistance genes was *Rivihabitans pingtungensis* which is a relatively newly described species mostly isolated from fresh water. Plasmids containing aminoglycoside resistance genes in multiple species might indicate active HGT, where the actual compatibility of these transfer activities may need further meta-transcriptomic sequencing study. This might suggest that the effort in eliminating the plasmids in the wastewater system can be a viable strategy in the control of aminoglycoside resistance dissemination (Wirth *et al.* 2026).

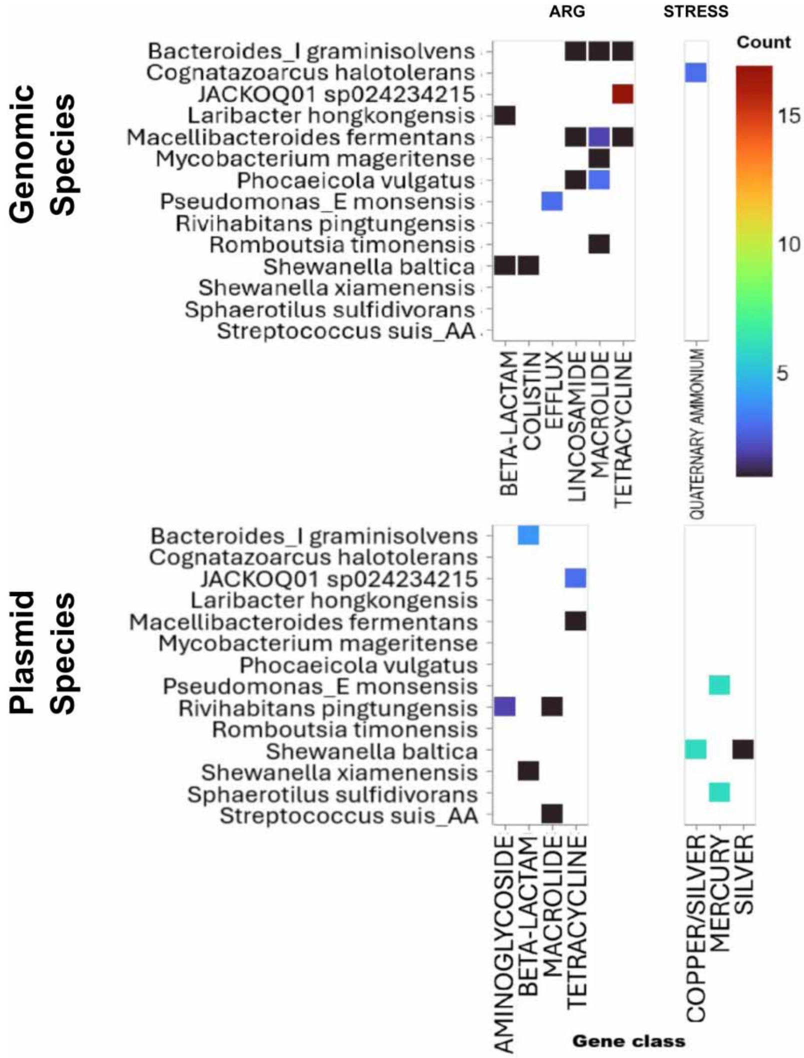


Figure 3 | Species-level fitness trait–host attribution heatmap. The horizontal axis provides gene classes grouped by type of fitness trait, and the vertical axis lists species with the gene element type details (plasmid versus genomic). Frequency displayed by color based on scale on the right.

Biocide resistance

Quaternary ammonium compounds (QACs) resistance was the only biocide resistance gene (within the stress group), as well as the only resistance gene carried by bacteriophages. QACs were recommended by the U.S. EPA to inactivate the SARS-CoV-2 (COVID-19) virus, and the use of these compounds has very likely increased during the pandemic (Hora *et al.* 2020). Previous study has shown the increased usage of QACs had affected the prevalence of quaternary ammonium resistance genes (Hora *et al.* 2020), and the results from this study indicated a potential spread of QACs resistance in *Aureliella* sp. via bacteriophages in South Florida according to GTDB (Ecogenomics 2025a). This information might urge us to re-evaluate the best practices for disinfectant usage which prevailed during the COVID-19 pandemic as well as the necessity for the removal of bacteriophage in WWTPs and new guidance in disinfection practice in clinical practice.

Potential applications of Hi-C metagenomic sequencing

The use of Hi-C metagenomic sequencing in understanding disease causes is relatively new, with applications found in understanding host microbe communities that harbor antimicrobial resistance for bovine dermal (Beyi *et al.* 2021a) and bovine respiratory (Beyi *et al.* 2021b) diseases. Drug resistance dynamics in opportunistic pathogens has been observed in human clinical microbiome samples (Ivanova *et al.* 2022). Combining this technology to evaluate wastewater, a repository of community level disease, provides the possibility of evaluating which host is carrying which ARG for a large human population. A study conducted in Denmark used wastewater to evaluate host associations of glycopeptide resistance with results showing that resistance was associated with *Enterococcus* and *Paenibacillus* species (Jensen *et al.* 2025). In the current study, a much broader view was taken by evaluating all metagenomically measurable ARGs and mapping them to potential bacterial hosts and identifying potential vectors of gene transfer. We envision that with further development the application of Hi-C metagenomic technology to analyze wastewater may provide information that would be useful for clinical practice, by connecting specific ARGs and bacterial hosts for clinically relevant illnesses that may circulate within communities.

Limitations

It is recognized that Hi-C metagenomic technology has its limitations, however. The information obtained through Hi-C represents the potential for HGT events (Calderón-Franco *et al.* 2023). The presence of vectors themselves does not guarantee a successful transfer and expression of ARGs. Confirmation of the expression of ARGs is recommended through further metatranscriptomic sequencing studies, such as through RNA-based sequencing. Also, Hi-C sequencing is technologically and bioinformatically intense, requiring greater laboratory and computational resources than conventional HTS approaches. To obtain more species-level connections between host, vectors, and fitness traits, much deeper sequencing would be needed, and we recommend a 90% threshold of completeness when evaluating sequencing quality. In addition, given the voluminous amount of data generated through Hi-C sequencing, applications of artificial intelligence should be considered for future applications.

CONCLUSION

This research employed Hi-C sequencing methods to not only identify ARGs, stress, and virulence genes, it also begins to identify the taxonomic hosts of these genes, which can provide insights as to the species transferring and receiving the genes, plus identifying the element types (vectors of plasmid or phage-associated) transferring the genes or whether the genes are found within the host's core genome. Results, therefore, provide proof-of-concept in the potential value of fitness trait–gene vectors–host attribution using Hi-C sequencing for samples of community wastewater. Results from this method of sample analysis can either be used to prove the results from previous studies or used to propose new hypotheses.

This study demonstrated that Hi-C sequencing provides results beyond those from traditional high-throughput metagenomic analysis. The results of Hi-C analysis in this study provided evidence of widespread resistance against ARGs in human pathogens and revealed potential HGT activities through plasmids, facilitating the spread of fitness traits in human pathogens, like resistance to aminoglycosides. The results demonstrated the spread of QAC resistance carried by both plasmids and bacteriophages.

With further advancements in Hi-C sequencing techniques, we envision that the host attribution approach can provide more detailed maps of localized host–vector–fitness trait interactions. Such details can benefit clinicians who prescribe therapeutics and administrators attempting to control the spread of ARG in the environment. With increased knowledge about the nature of infectious agents circulating in local community, the prescription of antibiotics can be more targeted to provide a higher probability of cures to infections; policy in public health can be made more effective to battle against antibiotic resistance dissemination; and advancements can be made in the understanding of the mechanisms underlying ARG spread with the details of host–fitness trait, or more specifically, host–ARG associations.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare the following potential conflict of interest: B.A., I.L., and Z.S. are affiliated with Phase Genomics, which provides commercial services related to the technology presented in this paper. This affiliation may be perceived as a potential source of bias. However, the research was conducted independently, and the findings presented in this manuscript are based solely on scientific evidence and analysis.

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